Bayesian quantile correction and synthesis for climate products

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QUESTION

Fast synthesis of corrected products

How can large retrospective product archives be used efficiently at an operational forecast origin to learn bias for each source, calibrate forecast ensembles across lead times, and synthesize predictive quantiles for the target process?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge. The framework learns dynamic corrections from USGS observations and retrospective ECMWF/GloFAS and NOAA/NWS products, then carries those corrections into the forecast window to synthesize calibrated predictive flow quantiles. This is needed because product families differ in lead time, update cycle, ensemble structure, and uncertainty representation.

USGS observations
15-min public gauge data; daily flow used for state updating and held-out verification.

ECMWF / GloFAS
daily issues; retrospective flow for bias correction; 51-member forecasts to 28 days.

NOAA / NWS
hourly issues; retrospective flow for bias correction; forecasts on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION

Rolling-origin holdout design

Frozen-origin validation. For each origin, source corrections are learned before forecasts are staged: the fit archive contains only USGS flow and product histories available by then. Future USGS is held out for scoring only. Main CRPS uses GloFAS-supported days 1–28; the NWS comparison uses days 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent Dynamic Quantile Linear Model (DQLM) lanes target $p_0 \in \{0.05, 0.20, \ldots, 0.95\}$. The fitted lanes are then assembled into one predictive distribution for scoring.

7	**$\leq$12,995**	**$\sim$2–4 h**
quantile lanes	USGS archive days	all lanes + synthesis

final-origin archive maximum; wall-clock on 64 cores, workflow/output-retention dependent.

Parallel quantile lanes
Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

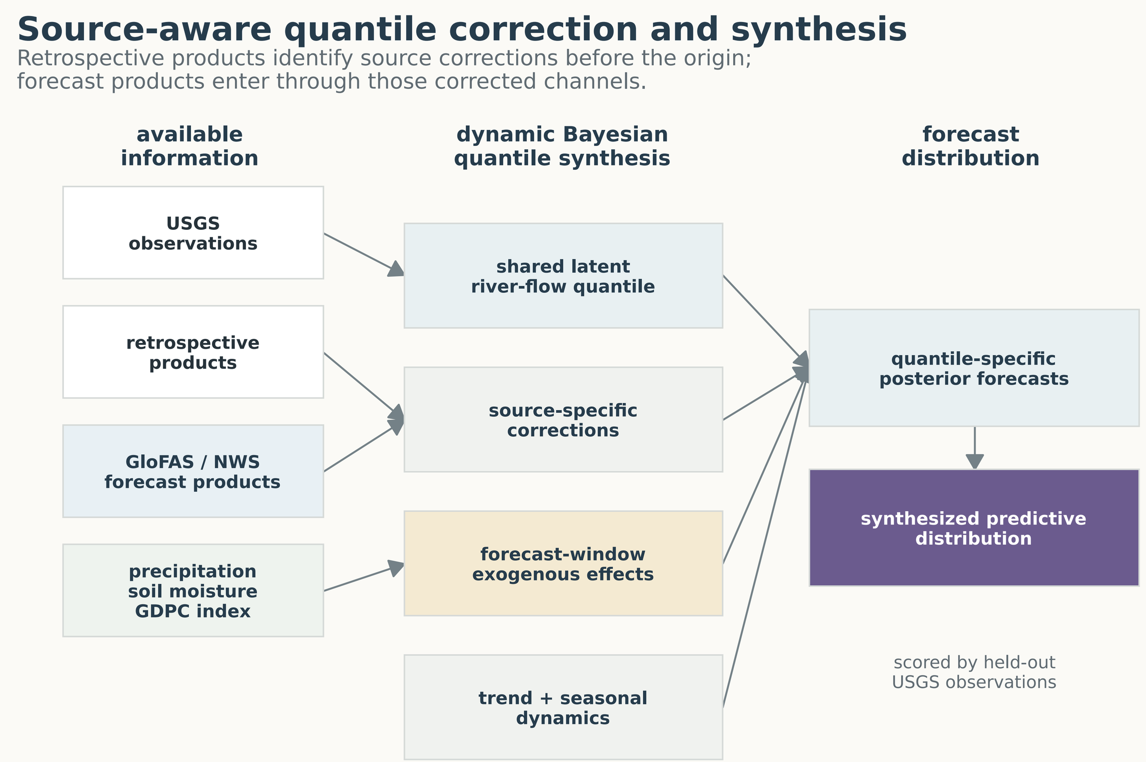
Posterior approximation
Mean-field variational Bayes (MFVB) avoids full Markov chain Monte Carlo (MCMC) while preserving Gaussian filtering and smoothing updates for latent state blocks.

Non-conjugate updates
Variational Bayes with Laplace–Delta (VB-LD) uses local mode, curvature, and Delta-method moments for non-conjugate extended asymmetric Laplace (exAL) scale and skewness terms.

State evolution
Component discounting controls trend, multi-period seasonality, source corrections, and exogenous-covariate effects.

MODEL

One state-space model for correction and synthesis



Correction before synthesis. Before the origin, observations and retrospective products identify the shared quantile state and source corrections. After the origin, issued forecast products and exogenous covariates update the same state to form calibrated predictive quantiles.

Expanded quantile state-space model
For each target quantile p_0 and forecast origin T, observed flow, retrospective product histories, and issued forecast members are linked through one latent river-flow quantile. Retrospective products identify source corrections before the origin; forecast products enter through those corrected channels.

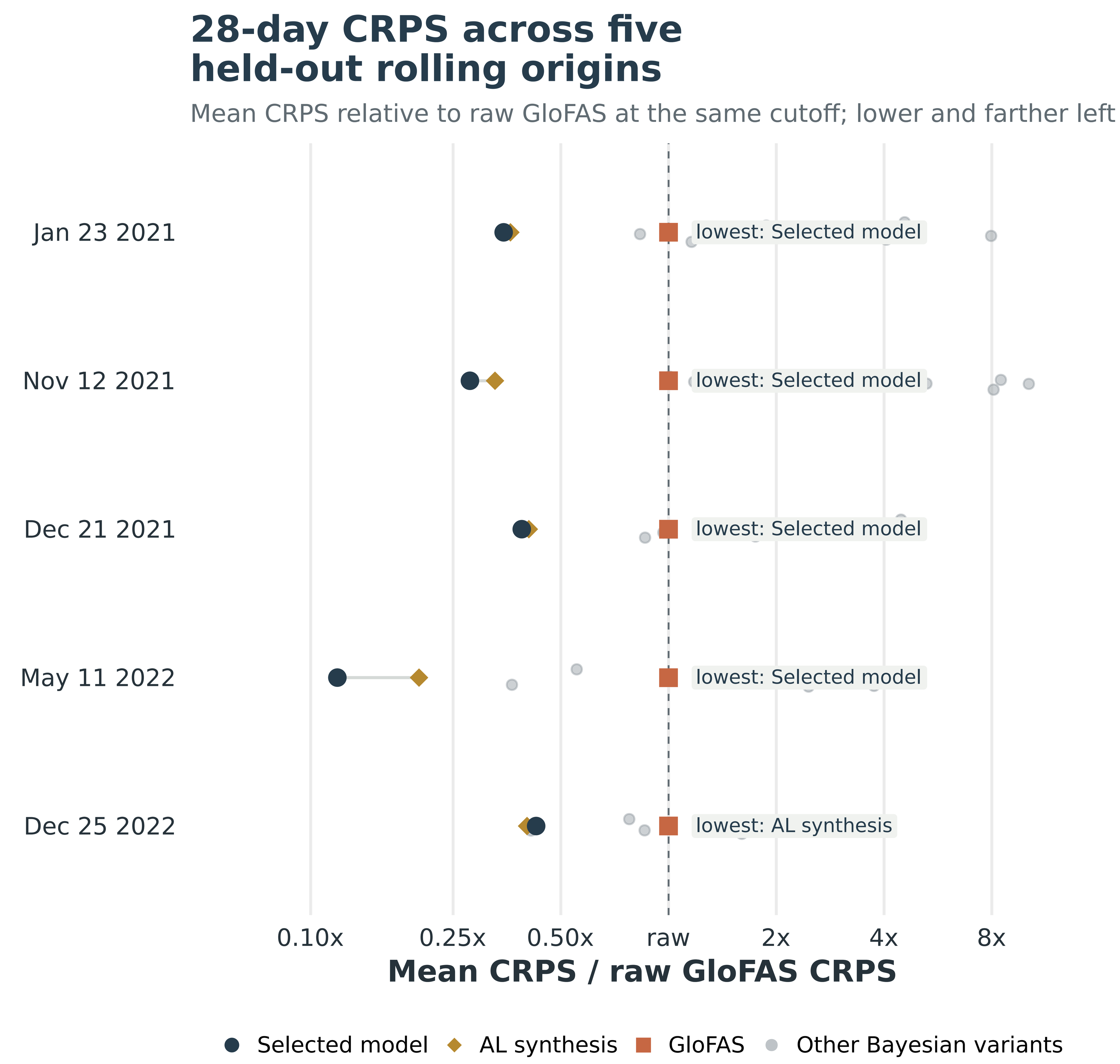
$$
\begin{array}{rlll}
\text{Observed measurements:} & y_t^o & \sim \mathrm{exAL}_{p_0}(\mathbf{F}_t'\boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o)\,, & 1 \le t \le T,\\
\text{Retrospective products:} & z_t^j & \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_t'(\boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j) + \zeta_t, \sigma^j, \gamma^j\right), & j \in \mathcal{J},\ 1 \le t \le T,\\
\text{Forecast products:} & y_T^{j,i}(k) & \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_{T+k}'(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j) + \zeta_{T+k}, \sigma^j, \gamma^j\right), & j \in \mathcal{J},\ i \in \mathcal{I}_j,\ k \in \mathcal{K}_j,\\
\text{Trend and seasonal states:} & \boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} & \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t^{\theta})\,, & 1 \le t \le T + K_{\max},\\
\text{Source correction states:} & \boldsymbol{\delta}_t^j \mid \boldsymbol{\delta}_{t-1}^j & \sim \mathcal{N}\left(\mathbf{G}_t\boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{\delta^j}\right), & j \in \mathcal{J},\ 1 \le t \le T + K_{\max},\\
\text{Exogenous adjustment:} & \zeta_t \mid \zeta_{t-1}, \boldsymbol{\psi}_{t-1} & \sim \mathcal{N}\left(\lambda\zeta_{t-1} + \mathbf{x}_t'\boldsymbol{\psi}_{t-1}, w_t^{\zeta}\right), & 1 \le t \le T + K_{\max},\\
\text{Covariate-effect states:} & \boldsymbol{\psi}_t \mid \boldsymbol{\psi}_{t-1} & \sim \mathcal{N}\left(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^{\psi}\right), & 1 \le t \le T + K_{\max}.
\end{array}
$$

exAL_{p_0} is the extended asymmetric Laplace likelihood; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, $K_{\max}$ the longest horizon, and $\mathbf{x}_t$ the exogenous covariates. $\boldsymbol{\theta}_t$ is the shared quantile state, $\boldsymbol{\delta}_t^j$ is the source correction/discrepancy state, and $(\zeta_t, \boldsymbol{\psi}_t)$ is the covariate-effect block. Priors use bounded truncated Student-t terms for γ^j, inverse-gamma terms for σ^j, Gaussian initial states, and component discounting for evolution variances.

MAIN EVIDENCE

28-day CRPS comparison

Held-out forecast-window distributional accuracy CRPS ↓



Grey points are the remaining Bayesian benchmark variants.

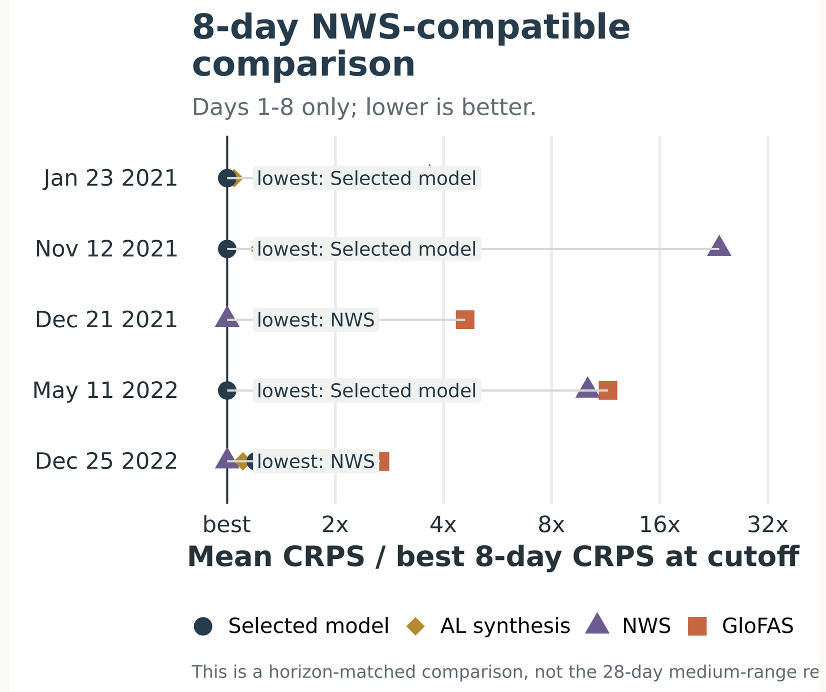
Use all five origins for final claims.

How to read this panel. Lower 28-day CRPS indicates better distributional accuracy over the held-out forecast window at each origin.

Values left of the raw GloFAS line indicate lower mean CRPS at the same origin.

HORIZON

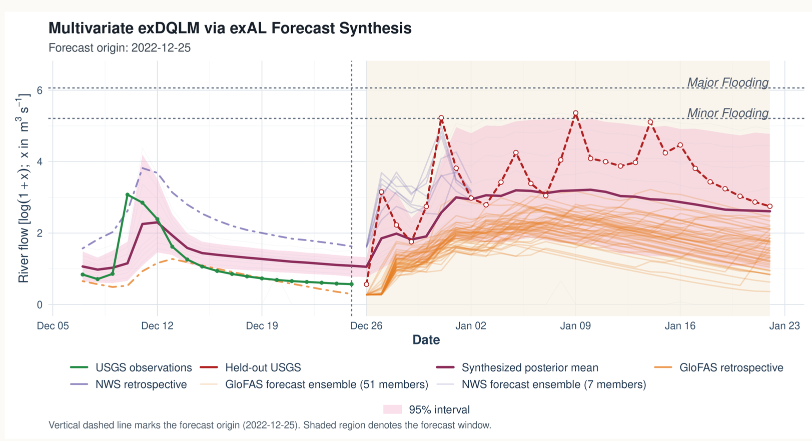
Horizon-matched NWS check



Horizon-matched evidence. NWS is available over common valid leads 1–8 in this evaluation. It is scored separately so the 28-day medium-range benchmark is not mixed with the NWS 1–8 day horizon.

EXAMPLE

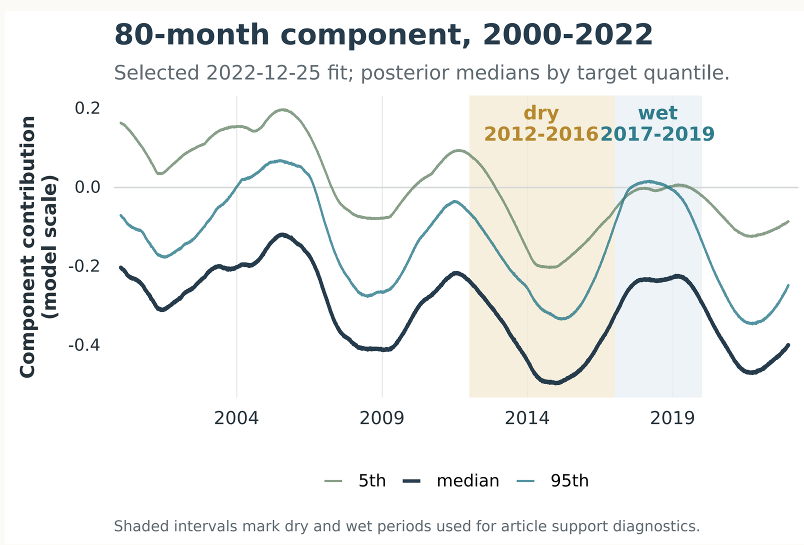
Representative forecast window



Illustrative, not aggregate validation. The panel shows how forecast inputs and posterior uncertainty evolve at the 2022-12-25 origin. It is illustrative; aggregate validation comes from the five-origin CRPS panel.

DYNAMICS

80-month component dynamics



More than forecast correction. The same state-space fit yields retrospective component diagnostics. The long-period harmonic tracks slow wet–dry structure while preserving quantile-specific variation.

INTERPRETATION

What the evidence supports

- Historically aligned retrospective products identify source corrections before the forecast window.
- Issued ensembles and exogenous covariates enter the state-space model directly, rather than being averaged after prediction.
- Observation-only quantile references have about 5–41x higher 28-day CRPS than the selected source-aware model across these origins.
- The five-window design is a targeted stress test, not a continuous hindcast; broader origin coverage is the next validation step.

Take-home. Learning source-specific corrections before synthesis can improve medium-range predictive distributions while preserving interpretable latent dynamics.